

Adaptive Classical Computer-Vision System for Fabric Defect Detection:

Week 2 Progress Report on Dataset Characterization

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Abstract—This report documents the Week 2 checkpoint of a course project to build an adaptive, classical computer-vision system for texture-based fabric defect detection and localization on the AITEX Fabric Image Database (AFID). We report only the quantitative outputs produced by running the current pipeline: dataset indexing and mask-matching statistics, image property checks, per-defect-code size distributions, sliding-window patch extraction counts, and per-descriptor texture-feature vector lengths (GLCM, LBP, Gabor, FFT). Feature extraction over the full patch set had not completed at the time of this report, so anomaly-scoring and detection results are deferred to the next checkpoint.

Index Terms—fabric defect detection, texture analysis, GLCM, local binary patterns, Gabor filters, AITEX AFID, classical computer vision

I. INTRODUCTION

Automated fabric defect inspection is a texture-analysis problem: a defect is a local deviation from an otherwise regular, periodic weave pattern. This project targets a classical (non-deep-learning) computer-vision pipeline that models normal fabric texture per fabric type and scores local patches by their deviation from that model, using the AITEX Fabric Image Database (AFID), which provides grayscale fabric images across multiple fabric types and defect categories along with pixel-level ground-truth defect masks. This report covers only the Week 2 deliverable: dataset auto-discovery, indexing, and characterization, run on Kaggle against the dataset mounted at `/kaggle/input/datasets/nexuswho/aitex-fabric-image-database`.

II. DATASET INDEXING

The pipeline auto-discovered the dataset root and classified every image file into `normal`, `defect`, or `mask` buckets by path keyword. Table I summarizes the resulting file counts.

TABLE I
FILE BUCKET COUNTS AFTER AUTO-DISCOVERY

Bucket	Count
Total image files	354
Defect-free (normal)	141
Defective	106
Ground-truth masks	107

Masks were matched to defective images by filename stem, with fallback suffixes (`_mask`, `-mask`, `_gt`). This matched

103 of 106 defective images (97.2%); the three unmatched files were `0044_019_04.png`, `0100_025_08.png`, and `0097_030_03.png` (Table II).

TABLE II
MASK-MATCHING OUTCOME

Metric	Value
Defective images	106
Matched masks	103
Match rate	97.2%

Defect-free images span 20 distinct fabric IDs, each with 7–8 samples (141 total). Defective images span 106 distinct fabric IDs, almost all with a single defective sample each.

A. Defect-Code Distribution

Table III shows the raw numeric defect-code distribution parsed from defective-image filenames (these codes have not yet been mapped to AITEX’s named defect categories).

TABLE III
RAW DEFECT-CODE DISTRIBUTION

Defect code	Count
02	38
03	31
01	12
04	10
00	7
06	6
08	1
05	1
Total	106

B. Image Properties

A sample of 30 images was checked for resolution, channel count, and data type (Table IV). All sampled images share a 256×4096 resolution and `uint8` dtype; channel count ranged from 1 to 4 (mean 1.2), indicating not all files are single-channel grayscale as assumed elsewhere in the pipeline.

C. Defect Size by Defect Code

Ground-truth mask area was computed as a percentage of total image area for each matched defective image and grouped by defect code (Table V).

TABLE IV
SAMPLED IMAGE PROPERTIES ($n = 30$)

Property	Value
Height	256 (constant)
Width	4096 (constant)
Data type	uint8 (constant)
Channels: mean	1.2
Channels: min / max	1 / 4

TABLE V
DEFECT AREA (% OF IMAGE AREA) BY DEFECT CODE

Code	n	Mean	Std	Min	Max
00	7	0.0622	0.0438	0.0158	0.1330
01	12	1.9211	6.4081	0.0030	22.2682
02	38	0.0415	0.0697	0.0009	0.2265
03	30	0.7022	2.5197	0.0000	13.6585
04	9	0.0503	0.1273	0.0017	0.3889
05	1	0.0273	—	0.0273	0.0273
06	6	0.0112	0.0125	0.0024	0.0311

III. PATCH EXTRACTION AND TEXTURE DESCRIPTORS

Sliding-window patches (64×64 , stride 32) were extracted from 40 sampled normal images and 40 sampled defective images. A patch was labeled defective if its overlap with the ground-truth mask reached at least 10% of its pixels. This produced 71,120 total patches: 70,645 normal and 475 defective, a defective-to-normal ratio of roughly 1:149 (Table VI).

TABLE VI
PATCH-LEVEL DATASET COMPOSITION

Quantity	Count
Total patches	71,120
Normal patches	70,645
Defective patches	475
Defective : normal ratio	1 : 149

Four texture descriptors were computed per patch and concatenated into a single feature vector: Gray-Level Co-occurrence Matrix (GLCM) statistics, uniform Local Binary Patterns (LBP), a Gabor filter bank, and FFT radial-band energy statistics. Table VII reports the per-descriptor and combined vector lengths as printed by the pipeline’s unit-test cells.

TABLE VII
PER-DESCRIPTOR AND COMBINED FEATURE-VECTOR LENGTHS

Descriptor	Length
GLCM	16
LBP	28
Gabor	16
FFT (radial-band)	8
Combined	68

At the time of this report, full-dataset combined feature extraction over all 71,120 patches had not finished executing.

Consequently, the per-fabric normal-texture model, anomaly scoring, and detection results are not yet available and are deferred to the next checkpoint.

IV. DISCUSSION

Three observations from the data collected so far are worth noting. First, the mask-matching step already resolves 97.2% of defective images automatically, leaving only 3 files for manual inspection. Second, defect-free and defective images cover largely non-overlapping sets of fabric IDs (20 vs. 106), which constrains how many fabric types can support a well-populated per-fabric model. Third, the sampled channel-count check (mean 1.2, max 4) indicates the dataset is not uniformly single-channel grayscale, which the pipeline should account for explicitly rather than assume.

V. CONCLUSION AND FUTURE WORK

This checkpoint establishes a reproducible dataset indexing and characterization stage for the AFID dataset, including mask-matching, image-property auditing, defect-size statistics by defect code, and patch extraction with a four-descriptor texture feature set. The immediate next step is completing combined feature extraction over all 71,120 patches, after which the per-fabric Mahalanobis anomaly model, score separability analysis, and heatmap-based localization can be evaluated against the ground-truth masks.

